

GUBBHOGEN, Sweden

WMO: 022300

Lat: 64.2172N

Lon: 15.5529E

Elev: 1017

StdP: 14.16

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.0	-7.5	-18.4	2.1	-12.9	-12.5	2.9	-7.3	18.7	33.5	16.3	32.7	1.5	290	0.659

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.4	76.6	61.3	72.9	60.3	69.2	58.6	64.0	72.7	61.9	70.1	59.9	67.1	5.5	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.7	82.2	67.0	58.6	76.1	65.2	56.6	70.8	63.3	29.7	72.8	28.2	70.1	26.8	67.5	70.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.9	13.6	11.7	-20.6	80.3	5.3	3.4	-24.4	82.7	-27.5	84.7	-30.5	86.6	-34.4	89.1		
(5)			-20.9	66.2	5.3	2.1	-24.7	67.7	-27.8	68.9	-30.7	70.0	-34.6	71.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	36.5	16.8	17.7	24.8	35.3	45.2	52.7	58.2	55.0	46.6	36.1	27.3	21.2	
	DBStd	16.56	12.32	12.09	9.14	6.06	6.79	6.20	5.75	5.29	5.32	6.57	8.40	11.68	
	HDD50	5528	1031	904	780	443	185	37	3	13	125	432	681	894	
	HDD65	10416	1496	1324	1245	892	615	370	223	314	551	896	1131	1359	
	CDD50	604	0	0	0	0	34	120	258	167	24	1	0	0	
	CDD65	18	0	0	0	0	0	2	13	3	0	0	0	0	
	CDH74	230	0	0	0	0	8	53	143	26	0	0	0	0	
	CDH80	22	0	0	0	0	0	1	20	1	0	0	0	0	
Wind	WSAvg	4.9	4.2	4.4	5.6	5.3	5.9	5.9	4.9	4.5	4.8	4.8	4.3	4.3	
Precipitation	PrecAvg	23.9	1.6	1.2	1.1	1.3	1.8	2.5	3.4	3.1	2.4	2.1	1.9	1.7	
	PrecMax	30.9	2.9	2.2	1.6	2.1	3.5	4.5	6.1	5.9	6.4	5.8	4.0	3.5	
	PrecMin	16.7	0.4	0.4	0.3	0.4	0.6	0.5	0.7	0.9	0.7	0.9	0.9	0.7	
	PrecStd	4.4	0.6	0.5	0.4	0.4	0.9	1.0	1.5	1.7	1.3	1.3	0.9	0.7	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.9	42.6	45.4	61.6	74.7	79.2	83.1	77.4	64.6	55.6	48.6	42.3	
		MCWB	38.4	38.9	39.2	48.0	56.8	59.9	65.4	62.5	56.9	49.6	46.0	39.4	
	2%	DB	39.3	40.1	42.2	54.6	68.8	74.7	77.9	73.1	60.7	50.0	43.1	39.5	
		MCWB	36.5	36.8	37.5	44.3	53.6	58.7	62.9	61.4	54.7	46.7	40.9	36.8	
	5%	DB	36.1	37.2	39.9	49.4	64.0	69.4	74.1	69.1	57.7	47.8	39.7	37.5	
		MCWB	33.9	34.7	35.9	40.9	51.4	56.6	61.7	60.1	52.9	45.5	38.2	35.3	
	10%	DB	32.5	33.5	37.9	45.4	58.4	64.9	70.3	65.4	55.2	45.8	37.2	34.9	
		MCWB	31.1	31.3	34.3	39.0	48.8	54.4	60.2	58.4	51.2	43.5	36.0	33.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.9	38.9	40.4	48.6	58.3	63.4	67.6	65.6	58.5	50.7	46.2	39.7
			MCDB	41.7	42.1	44.0	60.2	70.4	73.2	76.4	72.0	62.4	53.8	48.5	41.9
2%		WB	36.6	37.0	38.2	44.9	55.5	60.3	65.0	63.0	55.7	47.6	41.3	37.1	
		MCDB	39.1	39.7	41.6	53.5	65.1	70.5	73.8	70.0	59.4	49.4	42.8	39.1	
5%		WB	34.0	34.9	36.3	42.0	52.6	57.9	63.0	61.1	53.8	45.8	38.3	35.3	
		MCDB	36.0	37.3	39.6	47.9	62.2	67.3	72.2	67.5	57.0	47.7	39.5	37.1	
10%		WB	31.2	31.7	34.4	39.7	49.5	55.5	61.0	58.9	52.0	43.6	35.9	33.3	
		MCDB	32.4	33.4	37.5	44.8	57.7	63.4	68.8	64.7	54.5	45.6	37.0	34.6	
Mean Daily Temperature Range		5% DB	MDBR	11.5	12.5	14.9	15.0	17.3	17.0	17.4	16.4	13.2	9.8	7.7	10.2
			MCDBR	12.1	11.9	14.1	23.1	26.8	23.8	23.7	22.6	17.2	12.2	9.2	9.8
	MCWBR		10.6	10.4	10.7	13.6	14.2	11.9	11.5	12.6	11.9	9.7	8.3	8.8	
	5% WB	MCDBR	11.8	11.6	13.5	20.5	24.1	21.6	21.1	19.8	15.1	11.4	9.1	9.9	
		MCWBR	10.6	10.2	10.6	12.8	13.7	11.8	11.3	11.7	11.2	9.6	8.3	8.9	
Clear-Sky Solar Irradiance	taub	0.194	0.263	0.292	0.327	0.332	0.329	0.342	0.331	0.304	0.287	0.345	N/A		
	taud	1.988	2.214	2.287	2.374	2.479	2.515	2.493	2.512	2.574	2.461	2.770	N/A		
	Ebn at Noon	122	209	254	270	280	283	275	268	254	205	118	N/A		
	Edh at Noon	8	18	25	30	30	29	29	26	20	14	7	N/A		
All-Sky Solar Radiation	RadAvg	43	207	634	1195	1482	1647	1519	1132	640	261	67	18		
	RadStd	4	35	72	100	127	124	173	107	74	35	10	1		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (41 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page